ANDALUSIA ACADEMY

ITIDA-ACCREDITED · INDUSTRY-ALIGNED TRACKS

ACADEMIC TRAINING PROGRAMS PORTFOLIO

DESIGNED EXCLUSIVELY FOR FINAL-YEAR UNIVERSITY STUDENTS

PROGRAM OVERVIEW

These programs are specifically designed for university students in their final academic terms who are preparing to enter the job market. Each track is **accredited in partnership with ITIDA** and Andalusia Academy — so you graduate with nationally recognized proof of professional skill.

5	560+	ITI	100%
PROGRAMS	TRAINING HOURS	ACCREDITED	JOB-READY OUTCOMES

Introduction

ITIDA-ACCREDITED PROGRAMS — IN PARTNERSHIP WITH ANDALUSIA ACADEMY

These training tracks are officially developed and approved in collaboration with ITI (Information Technology Institute) and Andalusia Academy — ensuring every program meets national professional standards and is recognized by employers across Egypt's technology sector.

Built for university students in their final academic terms, these intensive, hands-on programs exist for one purpose: to make you genuinely employable the moment you graduate.

The gap between what universities teach and what employers actually need has never been wider. Our programs were built to close that gap. Each track is co-designed with industry practitioners to reflect the real tools, workflows, and expectations of today's technology companies — giving final-year students practical, demonstrable skills and portfolio-ready deliverables before they even step into their first interview.

These programs complement your university education — they do not replace it. They take your academic foundation and translate it into industry-grade competency. By the end of each track, you will have completed a real capstone project, worked with professional tools used in the industry today, and built concrete proof of your ability to contribute from day one.

JOB MARKET READY

Curricula built around what employers actually hire for — current tools, practices, and real technical expectations in Egypt's tech market.

ITIDA ACCREDITATION

Programs developed and approved in partnership with ITIDA and Andalusia Academy — nationally recognized and employer-trusted certificates.

PROJECT-BASED LEARNING

Every track culminates in a capstone project — a real portfolio piece, not just a certificate, to show employers what you can actually build.

FINAL-YEAR FOCUS

Specifically structured for final-year students — timed to complete before or alongside graduation for maximum career impact.

TRACK 01 · ITIDA ACCREDITED

Software Testing

100 Hours

Quality Assurance is one of the most in-demand entry-level roles in Egypt's tech sector. This program goes beyond traditional QA — equipping final-year students with AI-driven, cloud-native, and enterprise-grade testing skills that reflect the modern software delivery landscape.

CURRICULUM MODULES

- Advanced AI & Agentic Testing
- Modern Automation & Infrastructure
- Deep-Tech & Performance Testing
- Specialized Domain Validation
- QA Strategy & Governance

TOOLS COVERED

Playwright · Cypress · Appium · Docker · GitHub Actions · k6 · JMeter · Pact · Axe-core · Lighthouse · BrowserStack

CURRICULUM DETAIL

- Agentic Test Generation — Natural Language to Playwright / Cypress
- Self-Healing Automation Scripts — AI-Driven Locator Recovery
- AI-Powered Defect Root-Cause Analysis — Log Intelligence & Pattern Detection
- Predictive Analytics for High-Risk Module Identification
- Shift-Left Integration — Testing in the CI/CD Pipeline via GitHub Actions
- Containerized Testing Environments — Dockerized Test Grids
- Cloud-Native Testing — AWS Device Farm / BrowserStack Orchestration
- Visual Regression Testing — Pixel-Perfect Comparison Tools
- Contract Testing for Microservices — Pact / Postman Collections
- Load & Scalability Profiling — Distributed JMeter / k6
- Security-First QA — Automated Dependency Scanning & DevSecOps
- Synthetic Data Generation for Privacy-Compliant Testing
- Accessibility (A11y) Automated Auditing — Lighthouse / Axe-core
- Mobile-Specific Automation — Appium / Flutter Integration Testing
- Cross-Browser & Cross-Platform Compatibility Matrixing
- API Reliability & Chaos Engineering Basics
- Risk-Based Testing (RBT) Frameworks
- Test Data Management (TDM) & Masking Strategies

CAPSTONE OUTCOME

An AI-augmented test suite and automation framework for a real-world application — demonstrating advanced QA engineering skills to employers, from agentic test generation to containerized CI/CD pipelines.

PROGRAM DELIVERABLES

AI Test Plan & Automation Repo
Agentic scripts, self-healing locators, and visual regression suite on GitHub

Defect Intelligence Report
AI-generated root-cause analysis log with triage workflow and risk heatmap

CI/CD Pipeline Configuration
GitHub Actions workflow with containerized test grid and cloud orchestration

ITI Completion Certificate
Nationally recognized upon successful completion

CAREER OUTCOME
Junior QA Engineer, Automation Engineer, DevSecOps QA roles across Egypt's tech companies and outsourcing firms.

TRACK 02 · ITIDA ACCREDITED

Flutter Development

120 Hours

Mobile development is Egypt's fastest-growing tech hiring area. This advanced track takes Flutter well beyond the basics — covering on-device AI, high-performance rendering, enterprise infrastructure, and professional DevOps to build genuinely junior-ready mobile engineers.

CURRICULUM MODULES

- Advanced AI & Intelligence
- High-Performance Engineering
- Enterprise-Grade Infrastructure
- Professional DevOps & Security
- System Architecture

TECH STACK

Flutter & Dart · Firebase · TFLite · Shorebird · PowerSync · Rive · WASM · GitHub Actions · Dart FFI · Rust Bridge

CURRICULUM DETAIL

- On-Device LLM Integration — AI Toolkit & AI Model Integration
- Vector Databases & Semantic Local Search
- Edge ML Inference — TFLite / Vision AI & other providers
- Natural Language UI — Generative UX Patterns
- WebAssembly (WASM) for High-Performance Web
- Impeller Engine Optimization & Shader Warmup
- Background Isolate Orchestration — Heavy Data Processing
- Rive State Machines — Interactive Micro-Animations
- Server-Driven UI — Dynamic Remote Layouts
- Offline-First Data Sync — PowerSync / Drift
- Over-the-Air (OTA) Updates — Shorebird Code Push
- Feature Flagging & Remote Config Management
- Automated Visual Regression — Golden Testing
- Hardware-Backed Security — Secure Enclave / Biometrics
- Advanced Observability — Perf-Tracing & Memory Profiling
- CI/CD Build Flavors — Dev / Staging / Prod Orchestration
- Native Interop — Dart FFI / Rust Bridge
- Micro-Frontend Architecture — Package-Based Modularization

CAPSTONE OUTCOME

A production-grade cross-platform mobile app featuring on-device AI, offline-first data sync, OTA update delivery, and a fully automated CI/CD pipeline — your strongest proof of junior-level Flutter engineering.

PROGRAM DELIVERABLES

Capstone App Repository
Full source code with README, architecture docs, and CI/CD configuration

AI Integration Showcase
On-device LLM or Edge ML feature with documentation and performance benchmarks

Release & OTA Runbook
Pre-release verification checklist, QA sign-off, and Shorebird deployment config

ITI Completion Certificate
Nationally recognized upon successful completion

CAREER OUTCOME
Junior Flutter Developer, Mobile AI Engineer roles at startups, agencies, and product companies across Egypt and regionally.

TRACK 03 · ITIDA ACCREDITED

Data Engineering

Data engineering turns raw information into reliable, secure, and usable data products. This track prepares students to design modern pipelines, work with distributed systems, and build scalable data platforms using industry-standard tools.

CURRICULUM MODULES

- Data Engineering Foundations & Modern Data Stack
- SQL, Python & Data Modeling
- ETL / ELT & Pipeline Design
- Distributed Systems, Spark & Streaming
- Cloud Platforms, Data Quality & DataOps

TOOLS & PLATFORMS

Python · SQL · Apache Airflow · Apache Spark · Kafka · dbt · Docker · Terraform · AWS / Azure / GCP

CURRICULUM DETAIL

- Foundations of Data Engineering - data lifecycle, ingestion, storage, processing, and serving
- Distributed Systems Fundamentals - CAP theorem, replication, sharding, fault tolerance, and consensus basics
- Advanced SQL - analytical queries, window functions, CTEs, indexing, partitioning, and optimization
- Python for Data Engineering - APIs, file formats, Pandas, packaging, and testable pipeline code
- Data Modeling & Database Design - relational, dimensional, star and snowflake schemas
- ETL vs. ELT & Pipeline Patterns - batch, micro-batch, streaming, CDC, and incremental loading
- Apache Airflow - DAGs, scheduling, operators, monitoring, alerting, and retries
- Apache Spark - DataFrames, Spark SQL, PySpark ETL, partitioning, tuning, and job monitoring
- Data Lakes & Lakehouse Architecture - Parquet, Delta Lake, Iceberg, Hudi, and medallion design
- Cloud Data Platforms - storage, warehouses, managed pipelines, and cost-aware architecture

CAPSTONE OUTCOME

An end-to-end data platform that ingests, transforms, validates, and serves data for analytics - with orchestration, documentation, monitoring, and a production-ready deployment workflow.

PROGRAM DELIVERABLES

Data Pipeline Repository

Complete source code, documentation, tests, and orchestration configuration

Architecture & Data Model

Documented schemas, lineage, platform decisions, and security considerations

Monitoring & Quality Report

Validation rules, alerts, data quality checks, and operational runbook

ITI Completion Certificate

Nationally recognized upon successful completion

CAREER OUTCOME

Junior Data Engineer, ETL Developer, Analytics Engineer, and Data Platform Engineer roles across technology, healthcare, banking, and enterprise organizations.

TRACK 03 · CURRICULUM DETAIL

Data Engineering - Curriculum Map

The curriculum moves from core engineering fundamentals to scalable batch, streaming, cloud, governance, and machine-learning data workflows. Students learn how the components of a modern data platform work together in production.

FOUNDATION & CORE ENGINEERING

01	Foundations of Data Engineering & the Modern Data Stack
02	Distributed Systems Fundamentals
03	Advanced SQL for Data Engineering
04	Python for Data Engineering
05	Data Modeling & Database Design
06	ETL vs. ELT & Pipeline Design Patterns
07	Data Orchestration with Apache Airflow
08	Big Data Processing with Apache Spark I
09	Big Data Processing with Apache Spark II
10	Data Lakes & Lakehouse Architecture

PLATFORMS & OPERATIONS

11	Cloud Data Platforms
12	Streaming Data & Real-Time Pipelines
13	Analytics Engineering with dbt
14	Master Data Management & Integration Patterns
15	Data Quality, Governance & Security
16	Healthcare Data Interoperability & Standards
17	DataOps, CI/CD & Infrastructure as Code
18	NoSQL & Alternative Data Stores
19	MLOps & Feature Pipelines for Data Engineers
20	Capstone Project & Certification Preparation

GRADUATE CAPABILITIES

Build Reliable Pipelines

Design batch and streaming workflows that are repeatable, testable, and fault tolerant.

Model & Serve Trusted Data

Create well-structured analytical models with lineage, quality controls, and documentation.

Operate Modern Platforms

Use orchestration, cloud services, containers, monitoring, and CI/CD practices.

Support Analytics & AI

Deliver clean, governed data products for BI, machine learning, and real-time use cases.

PROGRAM FOCUS

Practical engineering decisions: selecting the right storage, processing, orchestration, and governance patterns; building maintainable systems; and explaining trade-offs clearly to technical and business stakeholders.

CAREER OUTCOME

Graduates are prepared for junior data engineering roles that involve pipeline development, SQL and Python, cloud data services, data quality, orchestration, and platform operations.

TRACK 04 · ITIDA ACCREDITED

Full Stack Development — .NET & React

140 Hours

Full stack developers are among the most hired profiles in Egypt's software industry. This is the most comprehensive track in the portfolio — covering both backend (.NET) and frontend (React) with real deployment, making you a complete and immediately hireable developer.

CURRICULUM MODULES

- Web Fundamentals & REST Principles
- C# & .NET Core Development
- React 18 & TypeScript Frontend
- Authentication, Testing & Security
- Deployment — IIS & Azure

TECH STACK

.NET 8 · C# · React 18 · TypeScript · SQL Server · EF Core · Azure

CURRICULUM DETAIL

- Web Fundamentals — HTTP/HTTPS, REST principles, authentication methods
- C# & .NET Fundamentals — OOP, LINQ, async programming, and the .NET ecosystem
- ASP.NET Core Web API — controllers, middleware, routing, and versioning
- Database Design & EF Core — relational design, migrations, and repository pattern
- Authentication & Authorization — JWT, Identity, and role-based access control
- React Fundamentals — components, JSX, hooks, and the React lifecycle
- State Management — Redux Toolkit or Zustand, global vs local state
- Frontend Routing & Forms — React Router, form libraries, and validation
- API Integration & Error Handling — Axios, interceptors, and loading states
- Testing — API testing with xUnit, frontend testing with Jest + Testing Library
- Deployment — IIS configuration and Azure App Service overview
- Capstone Application — full-stack project from design to deployment

CAPSTONE OUTCOME

> A production-ready full-stack application with documented API, deployed backend, and polished React frontend — your strongest portfolio piece.

PROGRAM DELIVERABLES

Full-Stack Capstone Repo
Complete source code for frontend and backend with full documentation

API Documentation (Swagger)
Auto-generated and annotated OpenAPI spec for all endpoints

Deployment Runbook
Step-by-step guide for deploying to IIS and/or Azure App Service

ITI Completion Certificate
Nationally recognized upon successful completion

CAREER OUTCOME

Junior Full Stack Developer, .NET Developer, Frontend Developer roles at software houses and product companies.

TRACK 05 · ITIDA ACCREDITED

Data Science, AI & Machine Learning

100 Hours

AI and Machine Learning are reshaping every industry. This track goes beyond classical ML — equipping you with the skills to fine-tune LLMs, build RAG systems, deploy scalable inference APIs, and operate production AI pipelines. These are the skills employers in Egypt's fastest-growing tech sector are actively hiring for right now.

CURRICULUM MODULES

- Vector Databases & Semantic Search
- RAG Systems & LLM Fine-Tuning
- Agentic Workflows & AI Orchestration
- MLOps CI/CD & Observability
- Scalable Inference & Deployment

TOOLS & LIBRARIES

Python · scikit-learn · TensorFlow / Keras · LangChain · ONNX · FastAPI · BentoML · Ragas · MLflow · Docker · Airflow · Jupyter

CURRICULUM DETAIL

- Vector Databases & Semantic Search — applied retrieval for ML feature lookup and nearest-neighbour search
- Hybrid Data Connectors — integrating real-time and batch data sources into ML training pipelines
- LLM-Based Data Extraction & Annotation — using LLMs to generate labels, parse documents, and augment datasets
- Feature Stores & Data Lineage — managing features at scale with reproducibility and audit trails
- RAG (Retrieval-Augmented Generation) Systems — building and evaluating retrieval-augmented LLM architectures
- Agentic Workflows & AI Orchestration — designing multi-agent systems with tool use, memory, and planning
- Model Fine-Tuning — PEFT and LoRA for efficient domain adaptation of large language models
- Object-Oriented Design for ML Systems — structuring ML codebases for maintainability, testing, and scale
- Model Serialization — ONNX and TensorRT export for portable, hardware-optimized inference
- Quantization & Model Optimization — reducing model footprint via INT8/FP16 quantization and pruning
- Scalable Inference APIs — FastAPI and BentoML for deploying ML models as production services
- GPU-Accelerated Containerization — Docker and Kubernetes for reproducible, scalable ML environments
- MLOps CI/CD Pipelines — automating training, evaluation, versioning, and deployment workflows
- AI Observability & Drift Detection — monitoring predictions, input distributions, and model degradation
- LLM Evaluation Frameworks — systematic quality measurement using Ragas, G-Eval, and custom metrics
- Distributed Training & Infrastructure as Code — multi-GPU / multi-node training and cloud provisioning with Terraform
- LLM Cost & Guardrail Management — latency budgeting, token optimization, and safe output enforcement
- Automated ML Workflow Orchestration — Airflow, Prefect, and Metaflow for end-to-end ML pipeline automation

CAPSTONE OUTCOME

A trained, evaluated, and production-deployed ML or LLM system — from data pipeline and model fine-tuning through RAG architecture and inference API — with a live demo and a polished research presentation.

PROGRAM DELIVERABLES

End-to-End ML Pipeline Repo
Full source: data ingestion, feature engineering, training, and serving code

Model Evaluation & Observability Report
Metrics, drift analysis, Ragas/G-Eval results, and performance benchmarks

Deployed Inference API
FastAPI or BentoML service with monitoring hooks and guardrails

ITI Completion Certificate
Nationally recognized upon successful completion

CAREER OUTCOME
Junior Data Scientist, ML Engineer, AI Research Engineer roles at tech companies, research labs, and enterprises across Egypt and the region.

Partnership & Contact

FOR FINAL-YEAR UNIVERSITY STUDENTS

Do not leave university without industry-ready skills and an ITIDA-accredited certificate. These programs are your bridge from academic knowledge to a real job offer. Enroll before your final semester and graduate employment-ready.

We welcome academic institutions, faculty advisors, and university career offices interested in integrating our training programs into their student development and career readiness offerings. Our team is available to discuss program customization, scheduling, and partnership structures tailored to your institution.

CONTACT EMAIL

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